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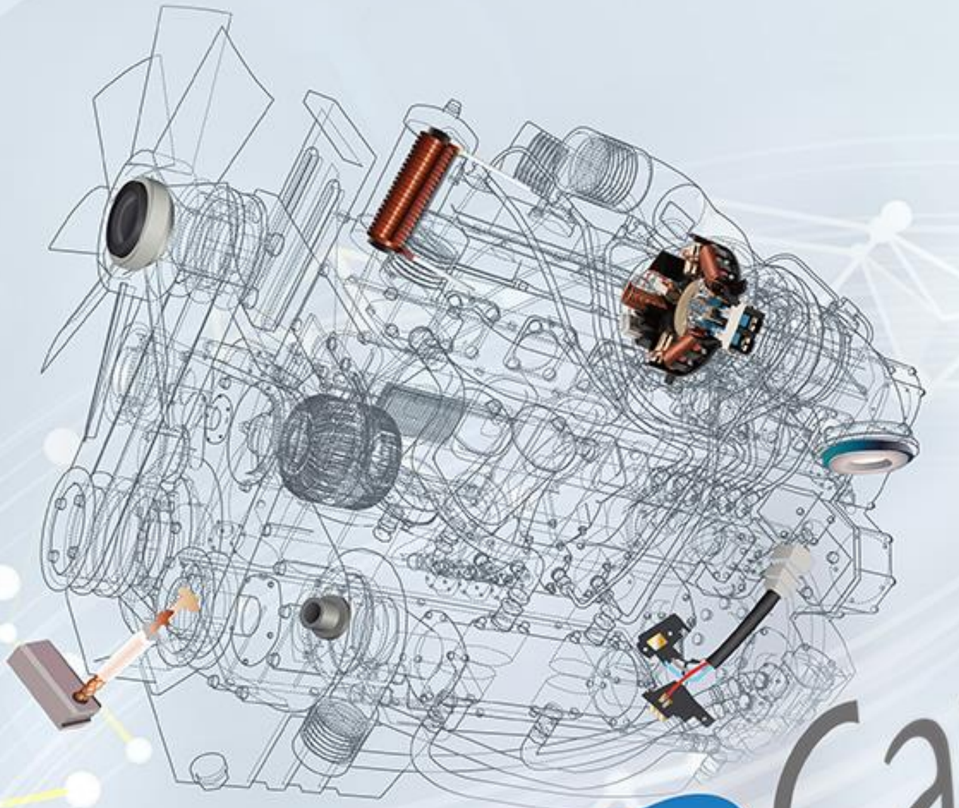

AVO Carbon GROUP

AVO Carbon Kunshan

Part No.: SAM33-DK Choke

一、团队组建 Team

Team: D1：组建团队	Name 名字	Dept 部门	Phone 电话
Champion(Problem solving experts)	Allan Regiel	General manager	13328055879
Team Leader:团队领导:	Tim	质量经理-QM	18068087785
Team Members:团队成员:	Ding shaoyang	电感工艺-PE	13306265707
	Chen ying	质量主管	13375155662
	Job	质量工程师-QE	18962613563
	Sky Song	仓库物流/Warehouse	13812951643
	Zhu zhonghua	生产主管Operation supervisor	13328055803
	Sunny	运营经理	18962460389
	Apple	Project manager	13358058033

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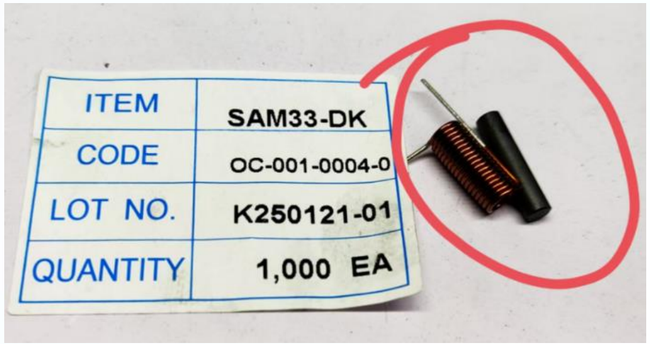
二、问题描述 Problem Description

D2 Problem Statement: D2 问题陈述：

产品料号Part No.:	SAM33-DK Choke	
零件号Part name:	Rod Choke	
发生日期Date of Detection：	2025/9/20	
发生地点Location of Detection：	Production line welding station 产线焊接工序	
缺陷数量 Defect quantity：	2PCS	
总数量Quantity manufactures	21000PCS	Batch No.： K250121-01
缺陷率Defect rate	0.0095%	
问题描述Problem description	Core Fallen Down	
对客户产线影响Failure effect at customer production line：	Affect the efficiency of customers' production and caused the motor can not work.影响客户的生产效率，造成电机无法工作	
0公里影响Failure effect at OEM （0KM）	Caused the motor can not work.	
最终客户影响Fialure effect on final custoemer （field）	Caused the motor can not work.	
发生频次How often does the problem occur?	1 times/ 2 Months	
历史发生频次What is the problem history	Occurred multiple times	
初步风险评估Preliminary risk assessment	Affect customer Assembly	End of D2： 20250930



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三围堵措施 Containment actions

产品遏制范围 Product Containment scope									
Part No.	区域 Location	措施Actions	责任人 Responsible	日期Date	控制方法 Control method	有效性 Effectiveness	总计围堵数量 Containment Quantity	总计检查/返工的数量 Total Quantity	不良数量Defect Quantity
	供应商库存 At Supplier	NA	——	——	——	——	——	——	——
	供应商第三方物流 Supplier TPL Third party Logistic	NA	——	——	——	——	——	——	——
	进货检验 Incoming Station	NA	——	——	——	——	——	——	——
	仓库 Warehouse Stock	Arrange operator sorting	Zhu Zhonghua	2025/9/21	放大镜下目视	100%	252800	252800	3pcs（疑似不良）
	制程库存 WIP-stock(Coil)	Arrange operator sorting	Zhu Zhonghua	2025/9/21	放大镜下目视	100%	0	0	0
	发运途中 In Transit	Arrange operator sorting	Job	2025/9/21	放大镜下目视	100%	0	0	0
	客户第三方物流 Customer TPL Third party Logistic	Arrange operator sorting	Job	2025/9/21	放大镜下目视	100%	0	0	0
	客户仓库 Customer Warehouse	Arrange operator sorting	Job	2025/9/21	放大镜下目视	100%	174000	174000	2 pcs（疑似不良）
	客户装配线上 At customer line	Arrange operator sorting	Job	2025/9/21	放大镜下目视	100%	0	0	0
	客户端仓库成品	Waiting for evaluation	Job	2025/9/21	放大镜下目视	100%	0	0	0
	市场端-0KM	NA	——	——	——	——	——	——	——
SAM33-DK	市场端-Field	NA	——	——	——	——	——	——	——
总数 Total									

区分可疑品的相关要求 Segregation of suspect products (pcs.):

1. 分选方法（如：目测、测量、Go/No go、Push test etc.） Sorting method (e.g. visual, test ,gage, Push test etc.).（提供分选、检查作业指导书）
2. 分选标准（合格/不合格的标准） Sort criteria (clear pass/fail standards)： Work instruction for Sorting
3. 合格的标记（如标记、标签、签字） I.D.method conforming (e.g. mark, tag, sign)
4. 不合格的标记（如：标记、标签、签字）I.D. method nonconforming(e.g.mark, tag,sign)

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围堵措施后可疑品验证 Verification of suspicious items after containment measures

AVO随线单品挑选已经完成252800 PCS，挑选出的疑似不良：3PCS，PPM:11，对挑选出的疑似品进行拆解，磁芯咬合状态良好，未发现类似磁芯脱落的产品，具体如下

The selection of individual items for the AVO line has been completed for 252,800 pieces. The suspected defective items identified: 3 pieces, PPM: 11. The suspected items have been disassembled. The magnetic core bonding condition is good, and no products with similar magnetic core detachment were found. :

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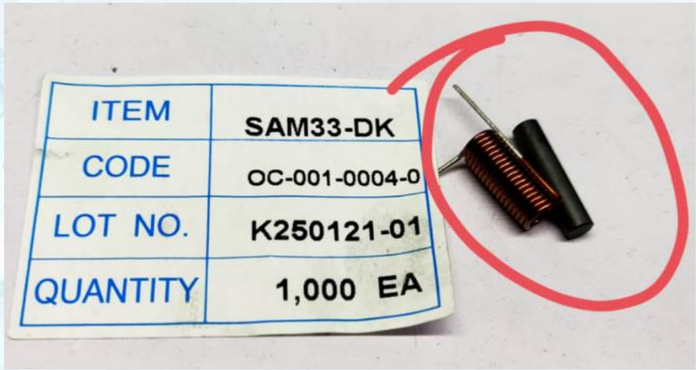
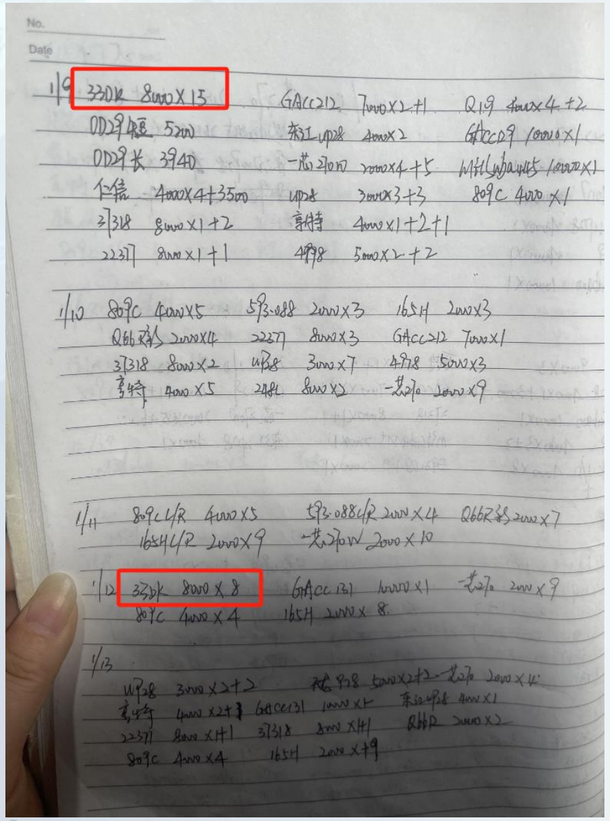
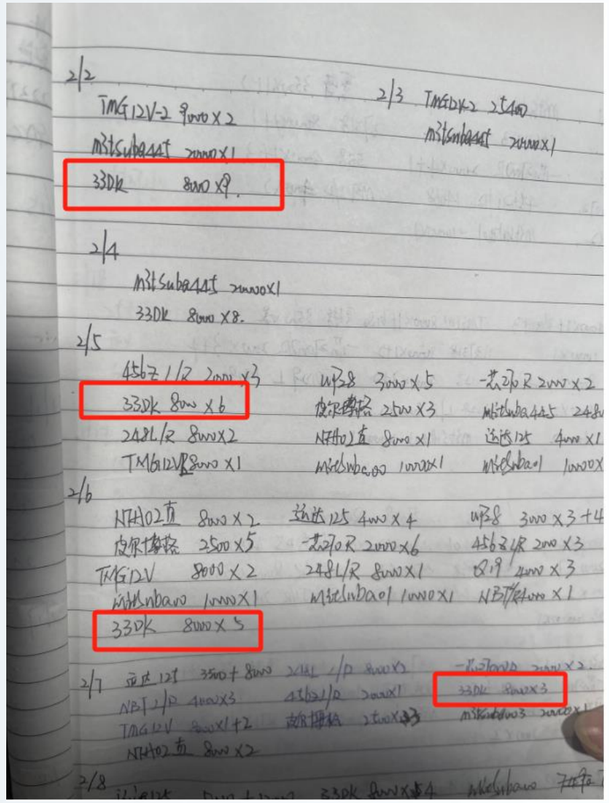
异常品生产批次分析Analysis of defective product production batches

异常包装批次



手工作业生产批次





总结：查询生产及包装记录，异常品为手工点胶生产批次

Summary: Review the production and packaging records. The abnormal products were from the manual dispensing production batch.

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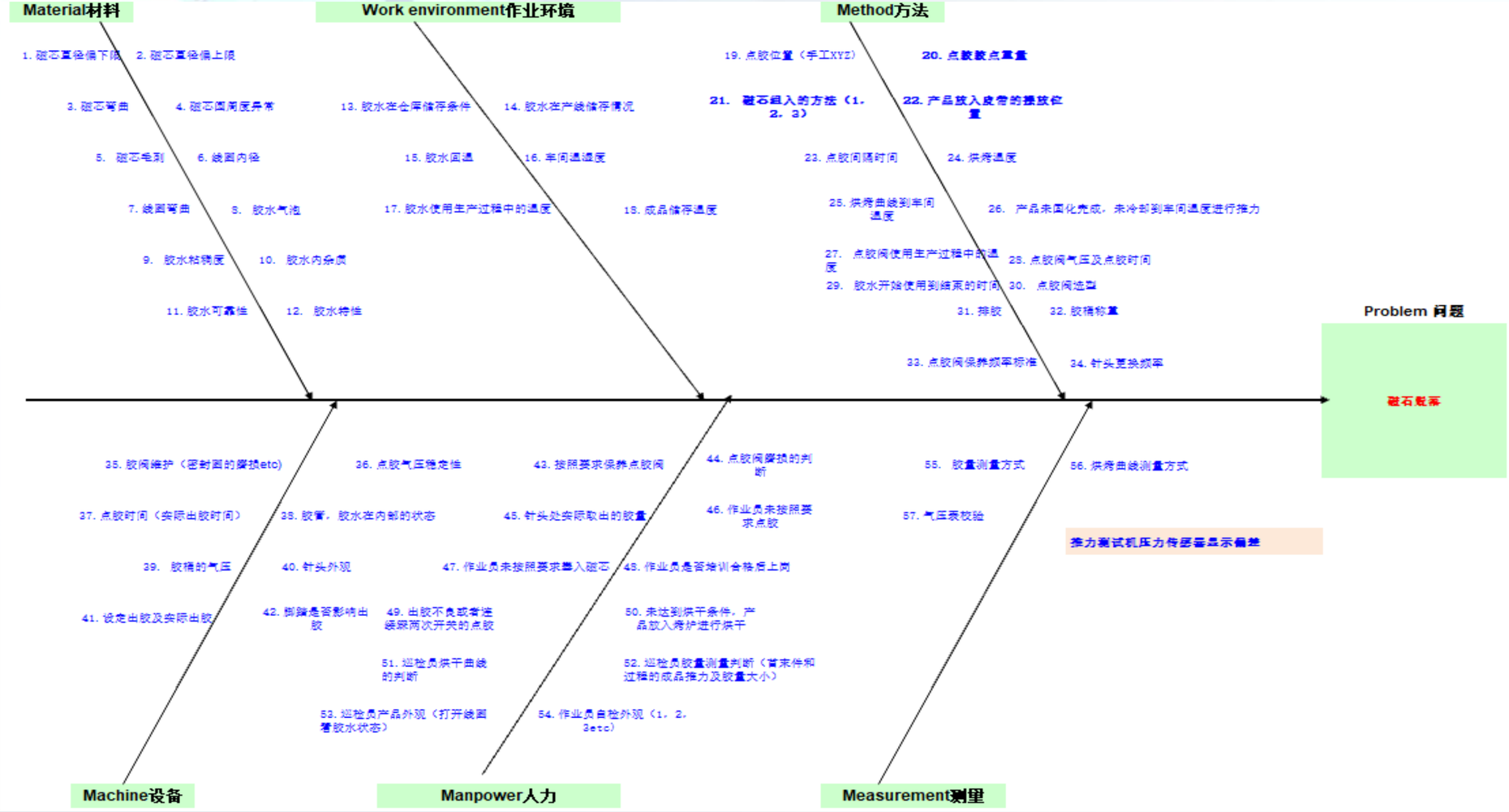
四、问题分析Root cause analysis：

1.根据SAM33-DK磁芯脱落异常脱落状况， 使用Ishikawa从人机料法环各方面对可能的原因分析，判定风险存在于如下工序:

Based on the abnormal of the gule on the core (SAM33-DK), Ishikawa is used to analyze potential causes, including Man, Machine, Material, Method, and Environment (4M1E). The risks existed in the following processes:



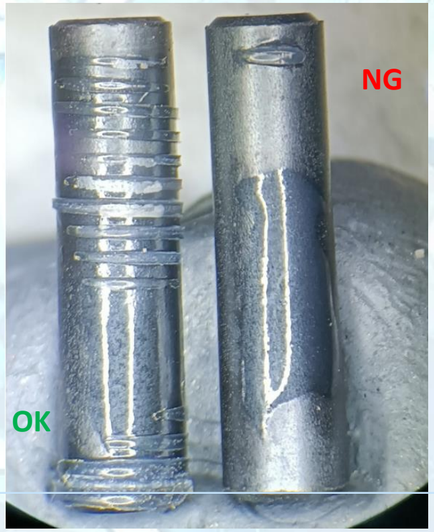
结论：根据以上分析潜在的风险工序为磁芯点胶和组入方式，胶量大小以及组装后产品在输送带中的位置

Conclusion: Based on the above analysis, the potential risky processes are the magnetic core dispensing and assembly methods, the amount of glue used, and the position of the product on the conveyor belt after assembly..

Supplier 8D report (EN-CN)-2024830-磁芯脱落-0919 的副本.xls

四、问题分析Root cause analysis：

2.分析缺陷件表面胶水和胶点咬合特征，根据缺陷件的缺陷特征，以及潜在的原因，分析人员可能的点胶操作方式以及放入烤炉中的状况，如下几种： Analyze the characteristics of the glue biting on the surface of the defective parts. Based on the defect characteristics, as well as potential causes, analyze the possible glue dispensing methods of the operator and the conditions of placing them in the oven, as follows:



缺陷件特征：

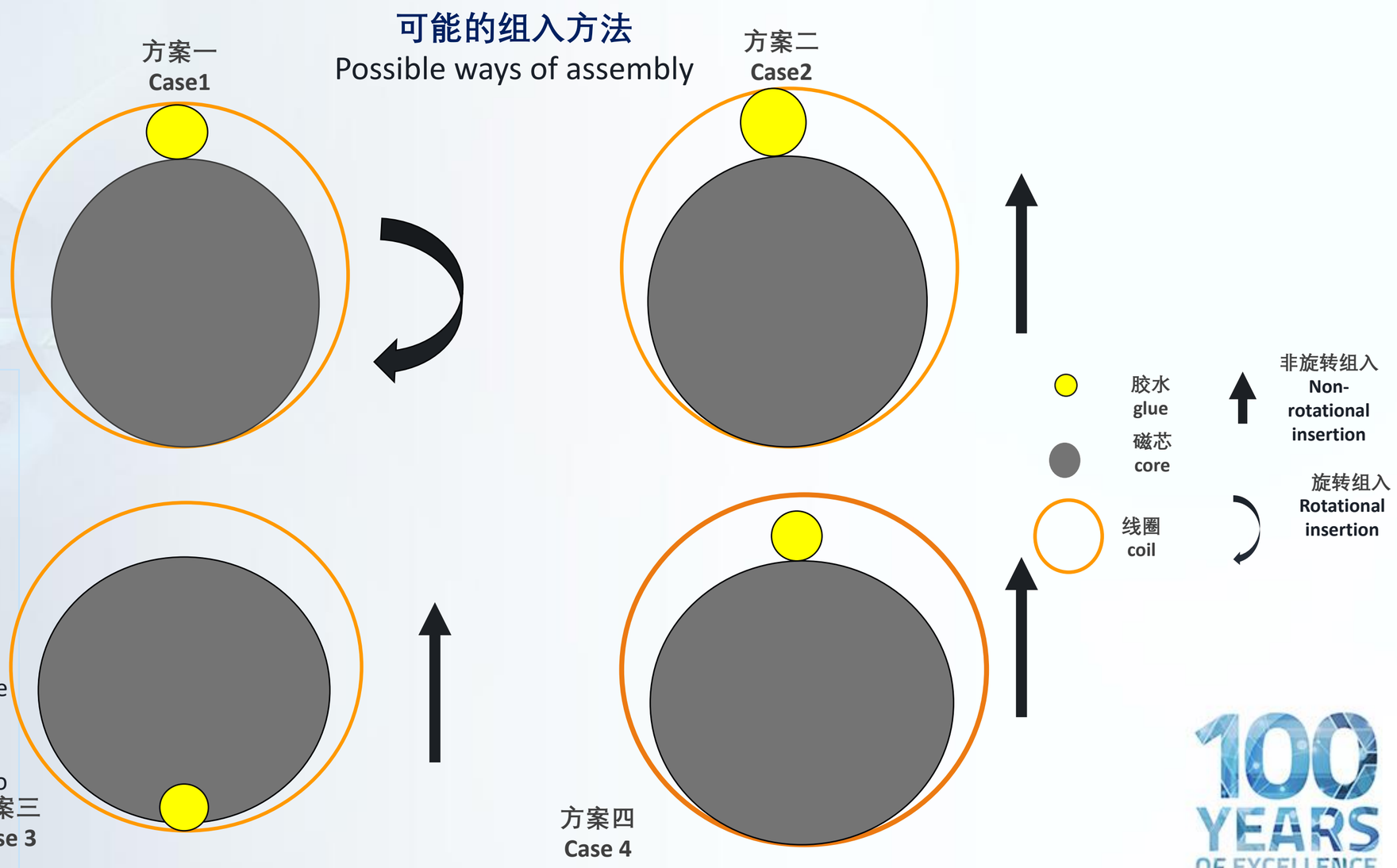
1. **磁芯上的胶量和正常件比起来明显偏少；**
2. **磁芯上的胶水和线圈咬合固化的印痕浅；**
3. **固化后的胶水集中在磁芯的一个部位，磁芯组入线圈时没有旋转。**

Defective product characteristics:

1.The amount of glue on the core is less compared to normal parts.

2.The biting & curing marks of the glue on the core and the coil are not deep.

3.The cured glue is concentrated in one area of the core, indicating that the core was not rotated when inserted into the coil.



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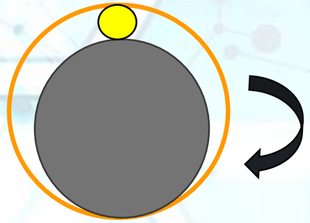
问题分析Root cause analysis：

3. 根据缺陷件的缺陷特征分析人员可能的点胶操作方式以及放入烤炉中后可能的结果：

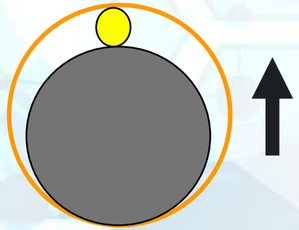
Based on the defect characteristics, the possible gluing methods used by operators and the potential results after placing them in the oven can be analyzed as follows:

方案一
Method 1



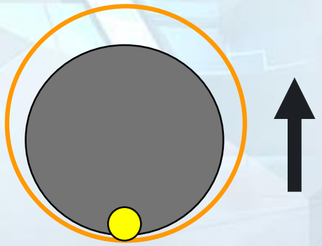
方案一：如果磁芯在组入线圈时旋转组入，磁芯上胶水会随着和线圈的磨蹭分布在磁芯周边；
if the core rotates when it is assembled into a coil, the glue on the core will be distributed around the core due to friction with the coil.

方案二
Method 2



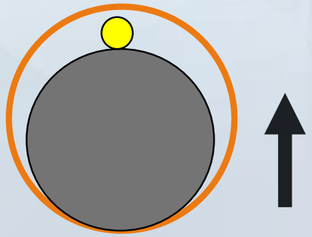
方案二：如果磁芯在组入线圈时未旋转组入，同时磁芯放在烤炉传送带上，磁芯因重力的作用下沉，可能磁芯和线圈咬合面少。
If the core does not rotate while being inserted into the coil, and at the same time, the core is placed on the conveyor belt of an oven, the core may sink under the influence of gravity, which could result in less contact between the core and the coil.

方案三
Method 3



方案三：如果磁芯在组入线圈时胶水在磁芯和线圈的下放，放置在烤炉传送带时因重力作用也是在下方，磁芯和线圈咬合痕迹多且明显。
If the glue is applied to the bottom of the core and the coil during assembly, then is placed on the conveyor belt of an oven with the same side down due to gravity, there will be more and more obvious bite marks between core and coil.

方案四
Method 4



方案四：如果磁芯在组入线圈时胶水在磁芯和线圈的上方，放置在烤炉传送带时因重力作用磁芯下沉，磁芯和线圈咬合痕迹浅显且不明显。
If the glue is applied to the top of the core and the coil during assembly, and the core sinks when placed on the conveyor belt of an oven due to gravity, the bite marks between the core and the coil will be shallow and not obvious.

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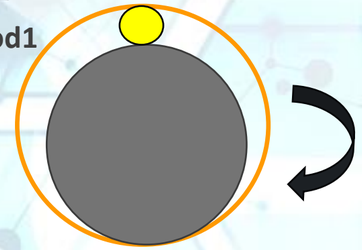
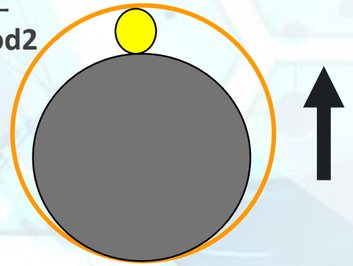
问题分析Root cause analysis：

4. 根据以上方案进行试验，试验结果如下/Based on the above method, experiments were conducted, and the results are as follows:：

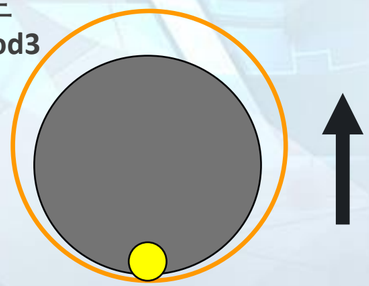
方案一
Method1



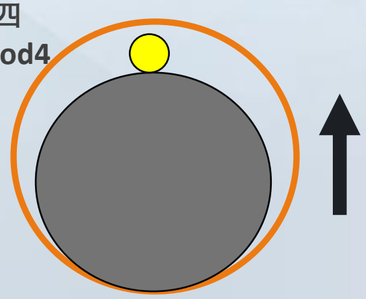
方案二
Method2



方案三
Method3



方案四
Method4



	胶量Glue weight			备注Note
	2.1mg	3.5mg	5.2mg	
方案一				
方案二				
方案三				
方案四				

结论：通过以上分析和实验可知，

1. 产品在方案4，胶量在下限&组装时磁芯没有旋转组入&烤炉皮带时胶点在上方，磁芯紧贴线圈底部，现象和异常品类似，以上要素叠加在一起导致了缺陷件的产生；

Conclusion: Method 4, with the minimum amount of glue & the magnetic core not rotating during assembly & the glue point being on the top when placed on the oven conveyor belt, causes the core to be tightly pressed against the bottom of the coil, similar to the phenomenon of the defective parts. The combination of these factors leads to the generation of defective parts;

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问题分析（流出） Root cause analysis escalation：

5. 产品只在过程中进行磁芯牢固度抽检，未做100%推力测试，

50	○			건조	건조기	1	건조기 온도				220±20℃	1회/일	계기판
						2	접착강도				1kg/30sec유지	2회/일	강도측정기
60	◇			자력검사	자력검사 컨베이어	1	빈 코일 혼입				빈 코일 선별	1회/일	육안
70				조립QA		1	외관				유해한 결함 없을 것	전수/LOT	육안
						2	치수	길이치수			16.0±0.2	n=5EA/LOT	V/C
								외경치수			5.25 MAX	n=5EA/LOT	V/C
								길이치수			8±1.5	n=5EA/LOT	V/C
								길이치수			5 MIN	n=5EA/LOT	V/C
								길이치수			6.0±1.0	n=5EA/LOT	V/C
								길이치수			4 MIN	n=5EA/LOT	V/C
						3	L값			▼	4.9μH±10% [at 1kHz]	n=5EA/LOT	LCR METER
80	○			포장		1					외부의 품질저하로 부터 보호 LOT 단위 완제품식별표 부착할것	BOX단위	육안, 전자저울
					전자저울	2	수량				6,000EA / 1BOX		
						1	외관				유해한 결함 없을 것	전수/LOT	육안
								길이치수			16.0±0.2	n=5EA/LOT	V/C
								외경치수			5.25 MAX	n=5EA/LOT	V/C

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五、手工点胶采取的措施The measures taken for manual dispensing of glue Corrective actions：

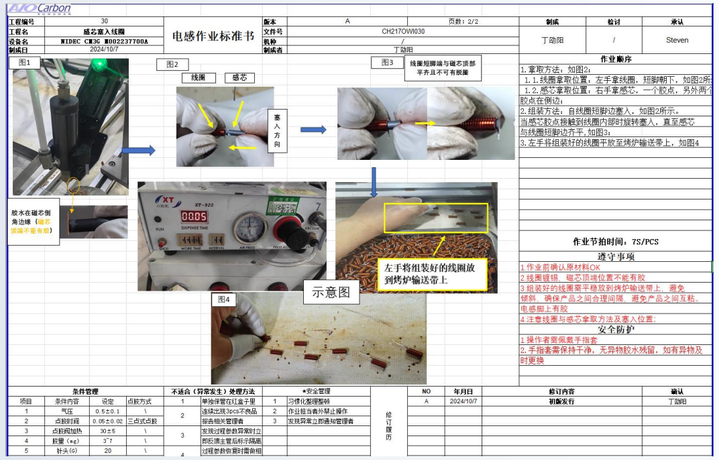
发生层面Occurrence：

1. **点胶方式由单点改为三点，对人员培训。责任人：Zhuzhu　时间：9月29日**
2. **更新点胶作业指导书(三点点胶，胶量调整为3-7mg 且旋转组入)。　Ding　10月9日**

The gluing method changed from single-point to three-point, and train OP. Responsible: Zhuzhu Date: Sep. 19th

Update WI(three-point gluing, with the glue weight 3-7mg and rotational as

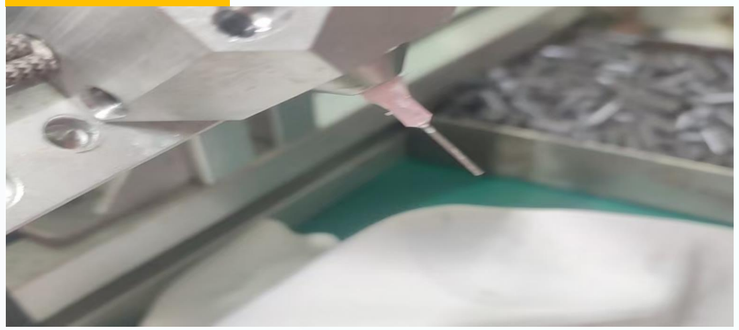
Updated SOP



备注：三点点胶，胶水出胶为激光和继电器共同控制，胶水出胶分散在磁芯端部三处，且旋转组入能够有效避免产品以上三要素叠加导致的异常，此工艺在突尼斯已经执行，至目前为止，突尼斯使用该工艺生产普通电感698000PCS，无磁芯脱落相关的不良。

Note: The glue is controlled by both a laser and a relay, distributing it to three locations at the end of the core. Rotational assembly effectively prevents the abnormality caused by the combination of the above three factors. This process is already implemented in Tunisia, where it has been used to produce 6,980,000 pieces without any core fallen off defects.

Before



After

五、手工点胶采取的措施The measures taken for manual dispensing of glue Corrective actions：

流出层面(Escape)：

推力测试后增加100%轻捏线圈检查是否存在磁石活动或者脱落现象（跟踪3个月）。

责任人：Sunny　　导入时间：9月26日

Add 100% manual inspection after push test to check for any caore movement or fallen off by lightly pinching the coil track 1 month.

Resp.: Sunny　　Date: Sep. 26th

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工程名称	外观检查	手推感芯 作业标准书	版本	1	制成	检讨	承认
工程名	感芯脱落检查		文件号	CH001OWI60R	Ding	Zhuzhu	Steven Li
产品名	22377 & 37318		工装名	/			
制成日	2024/9/4		制成者	Ding			

图1

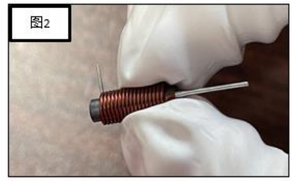
图2

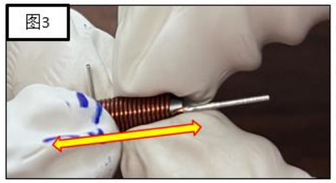
图3

图4

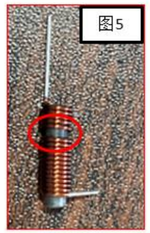
图5

图6

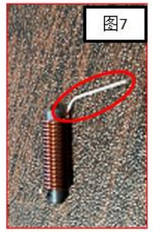
图7

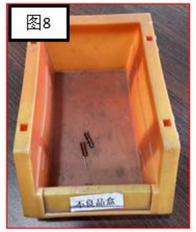

图8

作业顺序

1. 检查对象：两端无锁闭设计的电感（图1）；
2. 检查目的：检查电感感芯是否松脱；
3. 操作方法：
3.1. 右手拿取一只电感，两手指避开电感长脚，捏住约电感本体三分之二位置，如图2；
3.2　左手捏住电感三分一位置处，避开电感短脚，双手轻轻往两侧拉动，如图3；
3.3　如感芯未被拉出，则判定为合格品，集中放置于“检验合格区”，等待进入下一工序，如图4；
3.4　如果线圈被拉开，则判定为不合格，收集在不良品盒内，如图8。
4. 不良品状态：
4.1　线圈被拉开，如图5；
4.2　电感短脚变形，如图6；
4.3　电感长脚变形，如图7；

节拍时间：2000pcs/H

注意事项

1. 取放电感动作轻缓，避免感芯因碰撞而断裂；
2. 电感如不慎掉落，则直接废弃；

安全防护

1. 操作者需佩戴手指套

条件管理			不适合（异常发生）处理方法
项目	条件内容	设定	1 单独保管在红盒子里
1			2 连续出现3pcs不良品，报告相关管理者
2			★安全管理
3			1 习惯化整理整顿
4			2 作业担当者外禁止操作
5			3 发现异常立即通知管理者

修订履历	NO	年月日	修订内容	确认
	1	2024/9/4	初版发行	Ding

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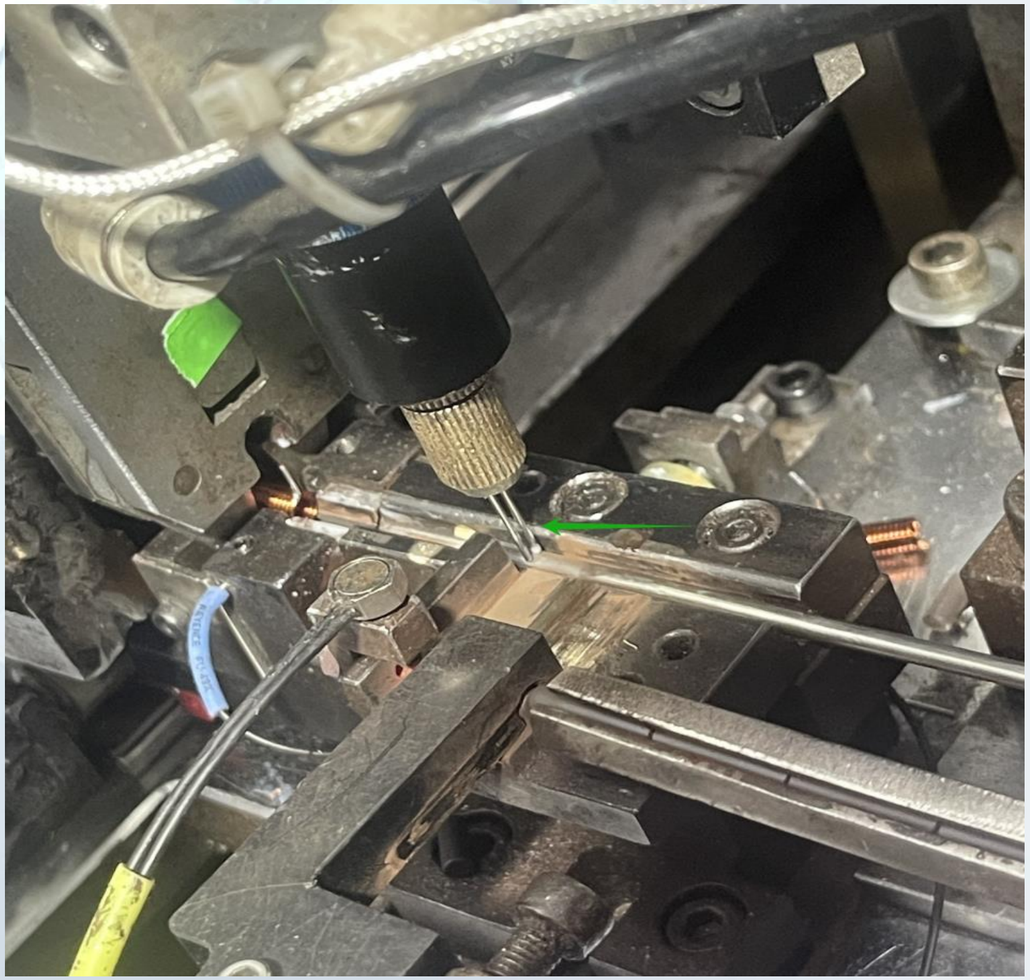
五、自动点胶机措施Automatic dispensing machine measures

发生层面Occurrence：

点胶针头改为双针头，增加涂胶面积及胶量，胶量控制在3-7mg.

The dispensing needle has been changed to a double needle, which increases the area and amount of glue application. The glue quantity is controlled within 3-7mg.



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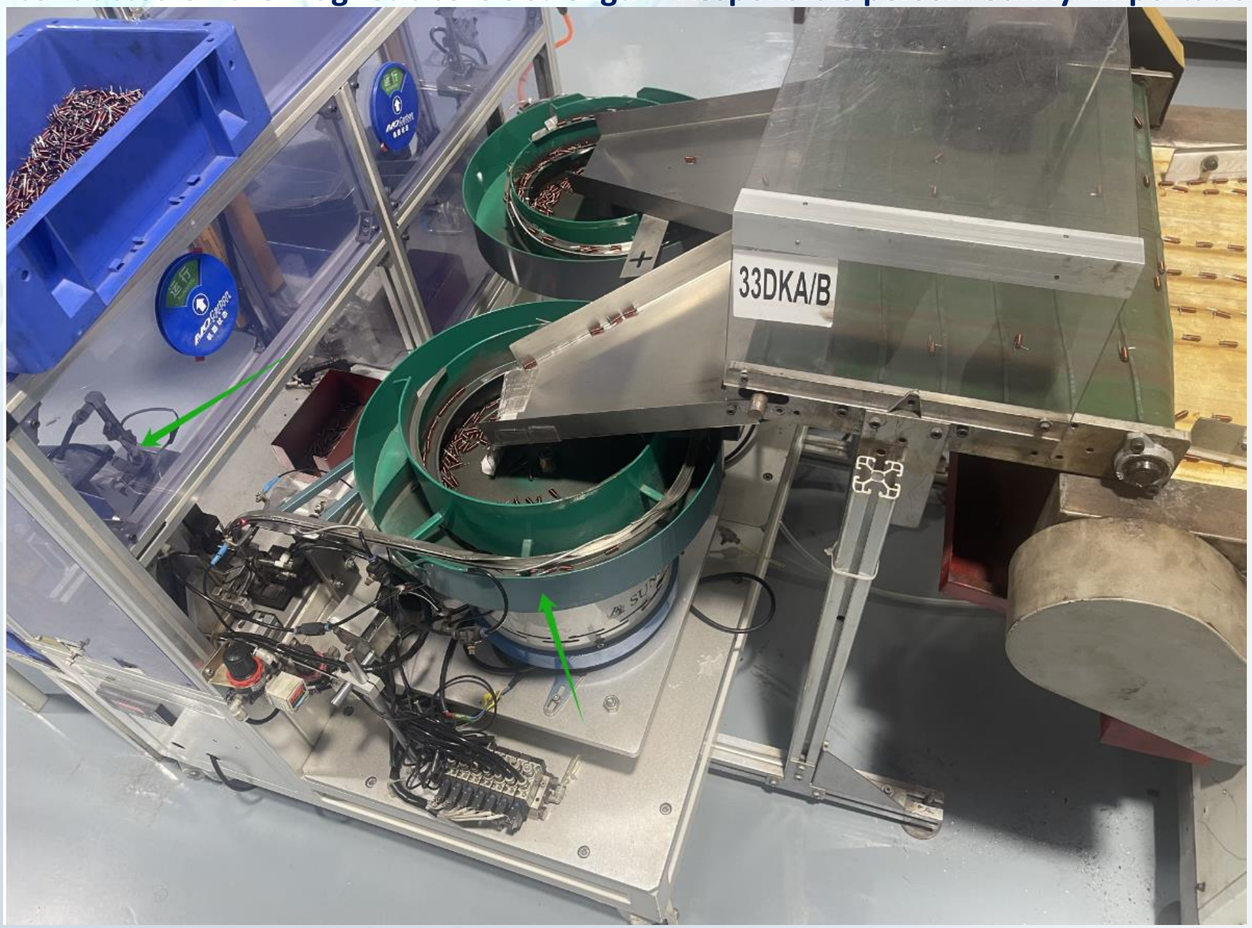
五、自动点胶机措施Automatic dispensing machine measures

流出层面(Escape)：

烘烤结束，由振动盘把成品电感直接导入推力测试机，对磁芯牢固度进行100%推力测试---责任人：Sunny　　导入时间：2月20日

The baking process is complete. The finished inductors are directly fed into the thrust testing machine by the vibrating plate, and a 100% thrust test is conducted on the magnetic core's strength - Responsible person: Sunny. Importation date: February 20th.



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